

DG INTERN HUB – CYBERSECURITY INTERNSHIP

WEEK 1 ASSIGNMENT REPORT: NETWORKING FUNDAMENTALS

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INTRODUCTION AND INTENSHP OVERVIEW

As part of the **Cybersecurity Internship at DG Interns Hub**, a series of theoretical and practical tasks were assigned to develop fundamental knowledge of computer networking. The tasks covered important concepts such as the **OSI Model, TCP/IP Model, network protocols, IP addressing, network troubleshooting commands, and network simulation using Cisco Packet Tracer**.

The purpose of these tasks was not only to understand networking concepts theoretically but also to gain practical experience by configuring and testing a small network environment.

The practical activities included using commands such as ping, ipconfig/ifconfig, and tracert, along with designing a basic network consisting of a **PC, switch, and router** using Cisco Packet Tracer.

The major objectives of the assigned tasks were:

- ❖ To understand the seven layers of the OSI Model.
- ❖ To understand the four layers of the TCP/IP Model.
- ❖ To compare the OSI and TCP/IP networking models.
- ❖ To study commonly used network protocols.
- ❖ To understand public and private IP addresses.
- ❖ To learn the basic concept of subnetting.
- ❖ To practice basic network troubleshooting commands.
- ❖ To understand how devices communicate within a network.
- ❖ To gain practical experience using Cisco Packet Tracer.
- ❖ To configure IP addresses on network devices.
- ❖ To test network connectivity using the ping command.

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8.	Learning Outcomes	Technical Skills & Problem faced
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TASK 1: OSI MODEL

The **OSI reference model** was developed by **International Standards Organization (ISO)**, which allows different communication systems to **communicate via standard protocols**. It gives a layered networking framework that conceptualizes how communications should take place among heterogeneous systems.

Layer No.	OSI Layer	Function	Technical Examples	Technical Examples
7	Application Layer	Provides network services directly to applications and enables users to access network resources.	HTTP, HTTPS, FTP, DNS, SMTP	When you open a website in Chrome, the browser uses application-layer protocols such as HTTP/HTTPS to communicate with the web server.
6	Presentation Layer	Translates, formats, encrypts/decrypts, and compresses data so that different systems can understand it.	Data encoding, encryption, JPEG, PNG, TLS-related data presentation	When you securely log in to a website, data may be encrypted before being transmitted and decrypted at the receiving end.
5	Session Layer	Establishes, manages, synchronizes, and terminates communication sessions between applications.	Session management, RPC, NetBIOS	During an online meeting, the communication session is established, maintained while participants communicate, and terminated when the meeting ends.
4	Transport Layer	Provides end-to-end communication, segmentation, flow control, error recovery, and reliable or fast delivery.	TCP, UDP	When downloading a file, TCP helps ensure that the data arrives reliably and in the correct order.
3	When downloading a file, TCP helps ensure that the data arrives reliably and in the correct order.	Provides logical addressing and determines the best path for packets between different networks.	IPv4, IPv6, ICMP, Routers	When you access a website hosted in another country, routers use IP addressing and routing information to forward packets toward the destination.
2	Data Link Layer	Provides node-to-node communication, framing, MAC addressing, and error detection within a local network.	Ethernet, MAC address, switches, Wi-Fi MAC	When your computer sends data to another device on the same Wi-Fi or LAN, MAC addresses are used for local delivery.
1	Physical Layer	Transmits raw binary bits through physical or wireless communication media.	Ethernet cables, fiber-optic cables, radio signals, connectors, hubs	When a PC is connected to a switch using an Ethernet cable, electrical signals carry the binary data between the devices.

The OSI Model provides a standardized and structured approach to understanding network communication. It simplifies networking concepts, makes troubleshooting easier, supports interoperability, helps in network design and protocol development, and provides a useful framework for analyzing cybersecurity threats at different network layers.

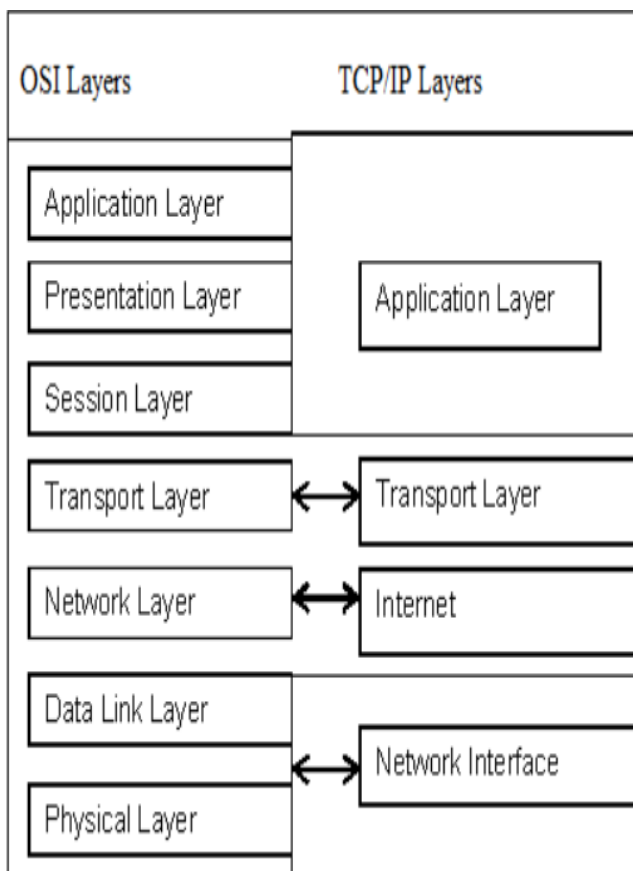
TASK 2: TCP/IP MODEL AND COMPARISON

Transmission Control Protocol/Internet Protocol (TCP/IP) model is a concise version of the OSI Model. The TCP/IP protocol suite was developed before the OSI model. But the layers in the TCP/IP protocol suite do not exactly match those in the OSI model.

It consist of 4 Layers:

- **Application Layer:** This layer provides network services to applications. The application layer uses HTTP, POP3, and SMTP protocols.
- **Transport Layer:** Responsible for communication between applications on different devices. It includes protocols such as transmission control protocol (TCP) and user datagram protocol (UDP).
- **Internet Layer:** Responsible for logical addressing and routing. IPv4, IPv6, ICMP, and routing protocols (among others) are the Internet Layer TCP/IP protocols.
- **Network Layer:** Responsible for communication over the physical/local network. It is concerned with the physical transmission of data.

Difference between TCP/IP and OSI Model



OSI	TCP/IP
Developed as a reference model	Developed around practical Internet protocols
Session and Presentation are separate	Included within Application layer
Physical and Data Link are separate	Combined into Network Access layer
Mainly used as a conceptual/reference model	Widely used for actual network communication
Protocol-independent reference framework	Closely associated with TCP/IP protocols

TASK 3: NETWORK PROTOCOLS

Network protocol is a collection of rules that governs how data is transmitted, received, and interpreted between devices on a network, regardless of their internal structure or design.

The assigned protocols were:

- **HTTP (Port no. 80)**
- **HTTPS (Port no. 443)**
- **FTP (Port no. 20/21)**
- **DNS (Port no. 53)**
- **DHCP (Port no. 67/68)**

HTTP (Hypertext Transfer Protocol): Used for transferring hypertext data between systems on the World Wide Web, it operates at the application layer.

- Functions on a client-server architecture (e.g., browser to web server).
- Standard protocol for loading web pages and transferring hypertext.
- Stateless, meaning it does not retain session information between requests.

HTTPS (Hypertext Transfer Protocol Secure): Provides secure communication between web browsers and servers by using encryption

- Uses SSL/TLS to secure data transmission.
- Protects user data from interception and tampering.
- Commonly used for secure web browsing.
- Ensures data confidentiality and integrity.
- Indicated by a padlock icon in web browsers.

FTP (File Transfer Protocol): Enables file transfer between a local system and a remote host

- Primary method for uploading and downloading files between local and remote systems.
- Built on the TCP/IP stack, ensuring that file data is delivered accurately and in the correct order.
- Extensively utilized for bulk file sharing and managing web server content.
- Requires a username and password for access, though it also supports "anonymous" sessions.

DNS (Domain Naming System): Translates human-readable domain names into IP addresses, allowing devices to locate and communicate with servers over a network.

- Maps domain names such as google.com to their corresponding IP addresses.
- Used when accessing websites, email services, cloud applications, and other Internet services.
- Uses a hierarchical system involving root, TLD, and authoritative DNS servers.
- Eliminates the need for users to remember numerical IP addresses.

DHCP (Dynamic Host Configuration Protocol): Automatically provides network configuration information to devices.

- Prevents IP address conflicts by managing and assigning available IP addresses.
- Allows devices to obtain network configuration automatically without manual IP configuration.
- Used in home, office, enterprise, and wireless networks to simplify network configuration.
- Automatically provides devices with an IP address, subnet mask, default gateway, and DNS server.

TASK 4: IP ADDRESSING

IP Address: An IP address is a logical numerical address assigned to a device connected to a network, used to identify the device and enable communication between different devices over an IP network.

There are two types of IP Address:- **IPv4 and IPv6 / Private IP and Public IP**

IPv4	IPv6
Uses 32-bit addresses	Uses 128-bit addresses
Provides approximately 4.3 billion addresses	Provides approximately 3.4×10^{38} addresses
Commonly uses DHCP for automatic configuration	Supports SLAAC and DHCPv6
Example: 192.168.1.10	Example: 2001:db8::1
Developed earlier and is still widely used	Developed to overcome IPv4 address exhaustion and support future network growth

Public IP	Private IP
Used to identify a device or network on the Internet .	Used to identify devices within a local/private network .
Usually assigned by an Internet Service Provider (ISP) .	Usually assigned by a router, DHCP server, or network administrator .
There is no single fixed range for all public IPv4 addresses.	IPv4 private ranges are 10.0.0.0/8, 172.16.0.0/12, and 192.168.0.0/16 .
Often used with NAT when multiple private devices share one public address.	Commonly uses NAT to access the Internet through a public IP.

Subnetting

Subnetting is the process of dividing a larger IP network into smaller logical networks called **subnets**. It helps in efficient IP address utilization, reduces network traffic, improves security, and makes network management easier.

Uses:

- Divides a large network into smaller networks.
- Reduces unnecessary broadcast traffic.
- Improves network performance.
- Makes network management easier.
- Provides better network security and isolation.
- Uses IP addresses more efficiently.

An IPv4 address consists of **32 bits**, divided into: **Network portion + Host portion**

Number of subnets: 2^n **Usable hosts per subnet:** $2^h - 2$

TASK 5: NETWORK COMMANDS

The practical networking task involved using basic network troubleshooting commands in Windows CMD / Terminal.

- **ping**
- **ipconfig**
- **ipconfig /all**
- **tracert**

- ❖ **PING:-** The ping command is used to test whether a destination is reachable over an IP network. It uses **ICMP Echo Request and Echo Reply** messages.

Purpose:

- Test network connectivity
- Check whether a host is reachable
- Measure approximate round-trip time
- Identify basic connectivity problems

```
C:\Users\jenif>ping google.com

Pinging google.com [192.178.211.113] with 32 bytes of data:
Reply from 192.178.211.113: bytes=32 time=7ms TTL=111
Reply from 192.178.211.113: bytes=32 time=6ms TTL=111
Reply from 192.178.211.113: bytes=32 time=10ms TTL=111
Reply from 192.178.211.113: bytes=32 time=6ms TTL=111

Ping statistics for 192.178.211.113:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 6ms, Maximum = 10ms, Average = 7ms
```

Observation:

The ping command was used to verify network connectivity between the local computer and the destination server. The command successfully resolved the domain name into an IP address through DNS and received ICMP Echo Replies from the destination. The response displayed the packet size, round-trip time, and TTL(Time To Live) value for each reply. The ping statistics provided the number of packets sent, received, and lost, allowing packet loss to be identified. The minimum, maximum, and average response times were also displayed, providing information about the latency and stability of the network connection. A successful response with 0% packet loss indicates that the destination was reachable during the test, although it does not guarantee that all application services on the destination are functioning.

- ❖ **IPCONFIG:-** It is a Windows Command Prompt command used to display the basic IP configuration information of the computer's network adapters.

Purpose:

- To check the computer's IP address.
- To view the subnet mask of the network.
- To identify the default gateway (router).
- To troubleshoot basic network configuration problems.

Observations:

The ipconfig command was executed to identify the basic network configuration of the computer. The output displayed the active network adapters along with their IPv4 address, subnet mask, and default gateway. These details help determine the device's network identity and the router used for communication with other networks. The command is useful for quickly troubleshooting incorrect or missing IP configuration.

```
C:\Users\jenif>ipconfig

Windows IP Configuration

Wireless LAN adapter Local Area Connection* 1:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 2:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix  . :
    Link-local IPv6 Address . . . . . : fe80::c3f5:bcef:fca7:33ef%16
    IPv4 Address. . . . . : 192.168.0.101
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 192.168.0.1
```

- ❖ **IPCONFIG /ALL:-** It is a Windows Command Prompt command used to display detailed network configuration information for all network adapters on a computer.

Purpose:

- To view detailed IP configuration of the computer.
- To check IP address, subnet mask, and default gateway.
- To find the MAC address and DNS server details.
- To troubleshoot DHCP and other network configuration problems.

Observation:

When ipconfig /all is executed, it displays detailed information about the available network adapters, such as IPv4/IPv6 address, subnet mask, default gateway, physical (MAC) address, DHCP status, DHCP server, and DNS servers. This information helps identify how the computer is configured on the network and is useful for diagnosing network connectivity.

```
C:\Users\jenif>ipconfigall
'ipconfigall' is not recognized as an internal or external command,
operable program or batch file.

C:\Users\jenif>ipconfig /all

Windows IP Configuration

Host Name . . . . . : Jenifa
Primary Dns Suffix . . . . . :
Node Type . . . . . : Hybrid
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No

Wireless LAN adapter Local Area Connection* 1:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : Microsoft Wi-Fi Direct Virtual Adapter
Physical Address. . . . . : AA-E2-91-DE-61-8E
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Local Area Connection* 2:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : Microsoft Wi-Fi Direct Virtual Adapter #2
Physical Address. . . . . : AA-E2-91-DE-71-9E
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Wi-Fi:

Connection-specific DNS Suffix . :
Description . . . . . : MediaTek Wi-Fi 6E MT7902 Wireless LAN Card
Physical Address. . . . . : A8-E2-91-DE-41-AE
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::c3f5:bcef:fca7:33ef%16(Preferred)
IPv4 Address. . . . . : 192.168.0.101(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Lease Obtained. . . . . : 04 September 2026 23:19:55
Lease Expires . . . . . : 05 September 2026 23:19:55
Default Gateway . . . . . : 192.168.0.1
DHCP Server . . . . . : 192.168.0.1
DHCPv6 IAID . . . . . : 162062993
DHCPv6 Client DUID. . . . . : 00-01-00-01-2F-75-14-9B-00-0F-C9-44-FE-50
DNS Servers . . . . . : 192.168.0.1
                        0.0.0.0
NetBIOS over Tcpip. . . . . : Enabled
```

- ❖ **TRACERT (Trace Route):** It is a Windows Command Prompt command used to trace the path taken by network packets from your computer to a destination and identify the routers or network hops along the way.

Purpose:

- To identify the path taken by packets to a destination.
- To find the number of network hops between your computer and the destination.
- To identify routers through which the packets pass.
- To troubleshoot network delays or connectivity problems.

Observation:

```
C:\Users\jenif>tracert google.com

Tracing route to google.com [192.178.211.139]
over a maximum of 30 hops:

  1  2 ms   1 ms   2 ms  192.168.0.1
  2  8 ms   4 ms   5 ms  10.2.0.1
  3 19 ms   5 ms   4 ms  172.25.5.25
  4  *      *      *      Request timed out.
  5 18 ms  10 ms   8 ms  26.188.100.175.mipl.com [175.100.188.26]
  6  9 ms  18 ms   6 ms  142.251.76.31
```

When tracert is executed, it displays a list of **network hops** between the local computer and the destination. Each hop generally represents a router or Layer 3 device along the path. The output shows the **hop number**, **response times**, and **IP address or hostname** of the responding device. Differences in response times can help identify where delays may be occurring. A * may appear when a hop does not respond to the probe or its response is filtered.

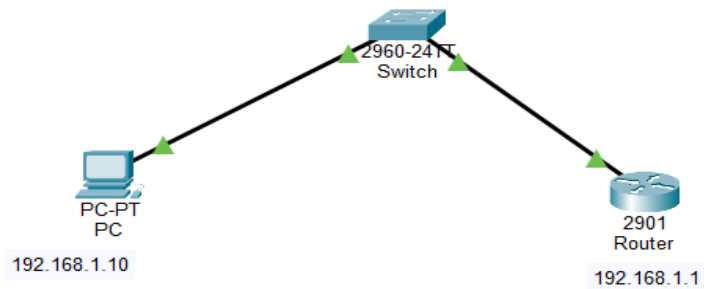
TASK 5: CISCO PACKET TRACER

Cisco Packet Tracer is a network simulation and learning tool that allows users to design, configure, and test network topologies without requiring physical networking equipment.

For this task, a simple network topology was created consisting of:

PC → Switch → Router

The purpose was to configure IP addresses and verify communication between the PC and router.



Procedure for the Above Configuration:

- Selected the appropriate devices — one PC, one switch, and one router.
- Placed the devices in the workspace and arranged them as PC → Switch → Router.
- Connected the devices using Copper Straight-Through cables.
- Connected the PC to the Switch using the appropriate Fast Ethernet ports.
- Connected the Switch to the Router using the appropriate Gigabit Ethernet interface.
- Configured the IP address of the PC as 192.168.1.10 (Static).
- Set the Subnet Mask to 255.255.255.0.
- Configured the router interface with the IP address 192.168.1.1.
- Set the PC's Default Gateway to 192.168.1.1.

Commands used to Configure the Router:

```
Router>enable
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface g0/0
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#no shutdown
```

Checking the Final Connectivity through the Ping Command:

```
Pinging 192.168.1.1 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time=10ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 10ms, Average = 2ms
```

This task provided practical understanding of network models, protocols, IP addressing, subnetting, and network troubleshooting. Commands such as ping, ipconfig /all were used to analyze connectivity, IP configuration, and packet paths. The PC–Switch–Router topology in Cisco Packet Tracer was also configured and tested using IP addressing and ICMP ping, strengthening practical networking and troubleshooting skills.

LEARNING OUTCOMES

- Gained a strong understanding of computer networking fundamentals, including the OSI and TCP/IP models and their role in network communication.
- Learned how different network protocols such as HTTP, HTTPS, FTP, DNS, and DHCP are used to provide communication and network services.
- Developed practical knowledge of IPv4 and IPv6 addressing, including public and private IP addresses, subnet masks, and basic subnetting concepts.
- Learned to analyze network configuration using `ipconfig` and `ipconfig /all`, including IP address, subnet mask, default gateway, DNS, DHCP, and MAC address information.
- Practiced using `ping` to verify network reachability, analyze packet loss, and observe response time and TTL values.
- Learned to use `tracert` to identify the path taken by packets and analyze individual network hops and response times.
- Gained hands-on experience with Cisco Packet Tracer by creating a PC–Switch–Router topology, assigning IP addresses, configuring the router, and testing connectivity.
- Improved basic network troubleshooting, configuration, and analytical skills, which are important for cybersecurity and network administration.

Problems faced:

- Initially faced difficulty in understanding the functions and responsibilities of the seven OSI layers and how they relate to actual network communication.
- Understanding IPv4 addressing, subnet masks, CIDR notation, and subnetting calculations required additional practice.
- Had difficulty interpreting some of the information displayed by `ipconfig /all`, particularly DHCP, DNS, and physical address details.
- While analyzing ping results, understanding response time, TTL, and packet loss required practical observation and comparison of different results.
- Initially found it challenging to interpret the hop numbers, IP addresses, and response times displayed by `tracert`.
- During the Cisco Packet Tracer implementation, minor issues were encountered while connecting devices and configuring IP addresses, subnet masks, and default gateways.
- Connectivity problems were resolved by systematically checking the cable connections, interface status, IP configuration, and gateway settings.
- Repeated practical testing helped in developing a better understanding of network troubleshooting and problem-solving techniques.

CONCLUSION

Overall Learning in This Week

This week, I developed a strong foundation in computer networking and basic network troubleshooting. I learned how the OSI and TCP/IP models organize network communication and understood the role of important protocols such as HTTP, HTTPS, FTP, DNS, and DHCP. I also gained knowledge of IPv4/IPv6 addressing, public and private IP addresses, and subnetting. Through practical activities, I learned to use ping, ipconfig /all, and tracert to analyze connectivity and network configuration. Creating a PC–Switch–Router topology in Cisco Packet Tracer provided hands-on experience in IP configuration, device connectivity, and troubleshooting. Overall, this week improved my technical knowledge, troubleshooting ability, and understanding of real-world networking concepts, which form an important foundation for cybersecurity.

Overall, this week successfully combined **networking theory with practical implementation**. The activities helped me understand how devices communicate across networks and how network problems can be identified and analyzed using standard tools. The practical experience gained through **command-line troubleshooting and Cisco Packet Tracer** strengthened my confidence in working with network infrastructure and provided a solid foundation for further learning in **Cybersecurity and Network Security**.